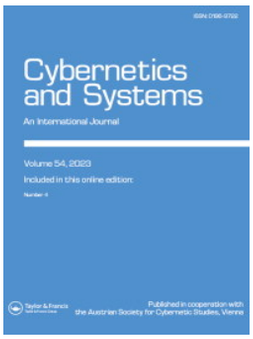

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RadScore: An Automated Technique to Measure Radicalness Score of Online Social Media Users

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ABSTRACT

Social media platforms provide effective mediums for expressing opinions and thoughts on several topics openly. This does protect our right to freedom of speech, however the enormous reach of social media makes it a potential tool for widespread radicalization among the youth, irrespective of the geographical and demographical boundaries. This necessitates the need to effectively identify the content which is a source of mass online radicalization. In order to curb the propagation, security agencies need an automatic radicalization detection mechanism for mining the huge volumes of social media content. In this article, we propose an approach for detecting online radicalized accounts and quantifying the degree to which these user accounts are propagating radical content. We propose to use three novel features, i.e., Similarity to domain, presence of radical content and sentiment to calculate the radicalness score for each online user. Our algorithm uses a CNN-LSTM-based technique to effectively differentiate between radical/non-radical content with an accuracy of 93%. Our empirical results show that radicalness scores for known radicalized websites are higher as compared to the non-radical users. We believe this is a first ever attempt at quantifying the level of radicalization of users using scientific methods which can be very helpful to national security agencies in tracking suspicious online users and stop the spread of anti-national content on social media.

KEYWORDS

CNN-LSTM; detection; radicalization; radicalness score; social media; social media analytics

Introduction

Radicalization is defined as "the action or process of embracing radical ideology that could lead to the commitment of terrorist acts" (Reinares et al. 2008). Modern day communication channels like social media serve as potent mediums for radical agents and anti-social elements to promote their ideologies and provoke the young minds. Social media channels like Twitter, Facebook, YouTube have become popular platforms for posting opinions and expressing emotions like anguish, resentment, and fear openly. When such negative emotions start resonating with the thoughts of

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the disillusioned youth, there is a huge possibility of minds getting radicalized. Thus, internet and these platforms can prove to be a big threat to the society at large. The RAND report (Von Behr et al. 2013) on Online Radicalization highlights the role of the internet in the radicalization of 15 terrorists and extremists in the UK. The report points out that online radicalization is on a rise among the youth because the internet has a much wider reach and has created more chances for people to get instigated. A similar trend can be seen in the conflict zones of Jammu and Kashmir. The number of young men and women getting radicalized via online mediums is alarming. There is an urgent need to design strategic and focused de-radicalization programs to bring back the radicalized youth.

Several international efforts have been undertaken to launch initiatives and programs for rehabilitation of people who have joined terrorist groups and performed violent acts. The aim of such programs should be to generate a counter narrative against radicalization and reintegrate radicalized people back into the society or at least dissuade them from violence and extremism. To make such initiatives successful, the first step is to develop mechanisms for detection of online radicalization of youth followed by sanitation procedures based on the degree of radicalization in an individual. There are only a few data driven/empirical studies available that have explored the role of technology in identification of youth radicalization on social media. Most studies are based on the post event analysis, i.e., after the incident of youth mobilization or radicalization has occurred. However, it is required that a system is designed which is capable of constantly monitoring the social media activities of targeted user accounts. This will help to prevent a potential harm to the national security. Artificial intelligence and machine learning techniques can be exploited to build such online monitoring systems that are capable of tracking social media handles, online accounts, and multimedia posted on social media. Any suspicious behavior or posts containing radicalized content should be flagged for further investigation. Groups, individuals, and network of people posting such content must be identified along with digital proofs.

However, the enormous size of social media data is impossible to be manually tracked or inspected. The variety and the volume of the online content is huge to be manually processed without error. This is where AI and machine learning can play an important part. Advances in technology like Big data, Natural language processing, and generation techniques have made it possible to automatically detect events of radicalization and extremism on online social media. Many studies have explored the use of ML-based classification algorithms for detecting hate speech and abusive language. There are several features in a sentence that can be exploited to indicate the presence of radicalization in text. Section related work of the article reviews these techniques in detail.

In this article, we have proposed an automatic radicalization detection and quantification approach. The proposed approach will help to detect and quantify the radicalized social media content into the levels of radicalness based on the thoughts being expressed in the post. Our research is a novel attempt to create a scoring mechanism for radicalized online accounts based on important factors like: (i) Similarity of text to domain (ii) Classification of text into Radical or non-radical (iii) Sentiment of the text. Such a scoring mechanism of radicalization can be used by Law enforcement agencies (LEA) to uncover radical propagation networks and carry out investigation against them. The radicalness scores can also be used by security agencies to effectively run the de-radicalization programs and sanitize the young deviated minds based on the degree of radicalness.

Related Work

Radicalization has been defined in several ways by many International agencies. A UNESCO report (Alava, Frau-Meigs, and Hassan 2017) defines radicalization as "anchoring oneself in one's knowledge, opinions, values and beliefs to determine one's behavior" while the Quilliam Foundation report (Hussain and Saltman 2014) describes it "as processes by which individuals and/or groups come to adopt extremist ideologies." The authors have emphasized on the fact that social media and Internet are the most popular mediums for the spread of radicalization. In the paper (Khosrokhavar 2015), the authors have mentioned violence as a key component of radicalization and explain the ways in which violence gets depicted in the behavior of a radicalized mind. In his opinion (Wagner et al. 2008), the author has described radicalization as a guerilla warfare where people with highly extremist mindset challenge the norms of society and the democratic setup. Thus, considering the available literature on radicalization, it can be inferred that radicalization is a process by which a "us vs. them" ideology gets developed into the human mind. The impact of such an ideology is so deep that the person resorts to violence to express oneself. This radicalization can be social, religious, political or criminal in nature. But fundamentally, the expression of radicalization is violent and full of hatred most of times.

There are several means by which radicalization can spread. The most modern tool for spread of radicalization in a large crowd without many restrictions is Social Media. In Thompson (2011), the authors have described how individuals and organizations make use of social media platforms to radicalize people for political advantages especially in the regions of Northern Africa and the Middle East. Since social media content is highly unregulated and it is very difficult to establish the identity of the

publisher, it is a popular choice among radical outfits for propagating their thoughts. The RAND report (Von Behr et al. 2013) has very effectively evaluated the role of internet in radicalizing minds by presenting 15 case studies of extremists and case studies in the UK. The authors have created 5 hypotheses for examining the role of Internet and social media in radicalization. The hypothesis states that Internet creates numerous chances for terrorist outfits to spread their word to a larger audience, it acts like an echo-chamber for radical thoughts, Internet accelerates the spread without any physical contact and many people are self-radicalized on the Internet. The 15 case studies provide valid evidence in favor of these hypothesis. Thus, proving that Internet is a powerful medium for spread of radicalization. In Mirchandani (2018), the authors have analyzed the difference between free speech and hate speech and how Social media has acted as a perpetrator in the modern times. The article explains hate speech and the subjects that come under the umbrella of hate speech in the Indian context. Examples of such subjects are Hindu vs. Muslim, Nationalist vs. Traitor, Rohingyas, Kashmir, etc.

Another interesting study (Zhao et al. 2013) suggests that the accounts spreading radicalization can operate in two modes: (i) Originators and (ii) Propagators.

- Originators: These are the nodes in a Social Media Network that are responsible for generating the first instance of a radicalized post. The thoughts expressed in these posts are very violent and have the capacity to steer the minds of the masses. To make the post more relatable and actionable, the originators use (i) contextual words like names of Politicians, Sensitive locations, (ii) mention events of importance (like bomb blasts, protests) and (iii) add violent or provocative messages to the post. The originators might be chiefs of a terrorist organization, religious leaders, or social activists.
- Propagators: These are the nodes in a social media network that are responsible for spreading the message generated by originators through online reposting, retweeting or forwarding the content. The propagators can communicate the original post or add their own opinion to the post and publish on social media. The propagators use creative means to spread the messages and usually have a huge, connected network of followers that blindly follow what the propagators express on social media. Propagators are usually local radicalized agents that have local connects and are popular amongst locals. They use their popularity to influence young minds. Propagators are the major channels through which the originators spread radicalization amongst the people.

The referred literature suggests that social media has facilitated the spread of radicalization and it is very important for the security agencies to develop counter strategies to curb the spread. There are several automated methods by which radicalization can be detected in text. Research works (Bermingham et al. 2009; Alghamdi and Selamat 2012; Wadhwa and Bhatia 2015; Koushik, Rajeswari, and Muthusamy 2019) use NLP-based techniques like Bag of words and TF-IDF for detecting topics and analyzing text on social media channels like Twitter, Facebook, Tumblr, and Youtube. In Ali, Mohammed, and Rajamani (2014), the authors have used Ontology-based information extraction technique and association rules to mine social media data and extract insights from it. The authors in Pujari et al. (2021) have used many machine learning methods to extract opinions from sentences on social media by counting the positive/negative polar words and finding dependencies in a sentence. Several research works (Agarwal and Sureka 2015; Hung, Jayasumana, and Bandara 2017) have made use of graph theory to identify radicalized groups and clusters in a social media network. Deep learning is a recent advancement in the domain of NLP and many researchers have used neural network-based techniques for detecting radicalization in social media content. The authors in Kaur, Saini, and Bansal (2019) have made use of LSTM networks for radical content detection and are able to achieve an accuracy of 85%. BERT, RNNs, Attention networks are also being explored by current researchers. The authors in Roy et al. (2020) have made use of Glove embeddings for designing a framework for hate speech detection. Particularly, in the paper (Fernandez, Asif, and Alani 2018), the authors have combined mathematical computations with social science theories to detect the presence of radicalization in a user account. The authors have compared the radicalization of 112 pro-ISIS with 112 "general" Twitter users to evaluate the difference in exposure. Various other techniques for text analytics include OP-D mathematical technique for classification (Mullick, Goyal, and Ganguly 2016), Summarization of text (Rudra et al. 2016a, 2016b), etc. All these techniques help to determine the behavior and polarity of text and the message conveyed by it with accuracy in a range of 70–90%. Recent studies like (Aldera et al. 2021a) have incorporated machine learning algorithms such as logistic regression, support vector machine, multinominal naïve Bayes, random forest, and BERT to detect extremist content in Arabic languages. The authors have concluded that SVMs and BERT models perform better than the other algorithms in text classification tasks in this domain. Another interesting study (Zahra et al. 2018) has used Multinomial naïve Bayes to characterize a user into extremist and non-extremist categories. The authors in Muhammad Asif et al. (2020) have suggested a techniques based on SVM and Naive Bayes which detects the sentiment present in multi lingual

sentences. The algorithm has been able to achieve a validation accuracy of 88%. To improve the performance of classification task, the authors in (Mashechkin et al. 2019) have suggested the use of Decision tree, Logistic regression, Random Forest, Light GBM, and XGboost algorithms. The article also studies the relationship among the communities in which presence of extremism is strong. Some notable review studies related to Extremism is (Aldera et al. 2021b; Gaikwad et al. 2021) in which the authors have presented a thorough analysis of different techniques for detection of extremism in textual data. It is evident from this study the sentiment analysis and semantic similarity are important factors for detecting extremism and radicalization in text. Several techniques make use of latest machine learning and deep learning libraries to detect radical content in online social media.

However, it can be seen that no individual technique is able to quantify the level of radicalization of an online user account. It is essential that the each account is scored according to its degree of radical activity and thus appropriate action can be take. Taking motivation from literature, we have incorporated three important text-based features in our proposed approach to quantify radicalization. These features are: (i) Similarity of text to domain, (ii) Classification of text into Radical or non-radical, (iii) Sentiment of the text. All the three features are important indicators of user intent and can help in quantifying the degree to which an online user account is radicalized. The following sections explain the proposed approach in detail.

Research Methodology

The objective of the current study is to design an efficacious scoring mechanism for estimating the severity of radicalization in a social media user. The scoring mechanism will help to quantify the degree of radical content being propagated by a user account into a large social media network. For this purpose, we have proposed a mathematical approach to calculate the radicalness score of the online user using three important text related features. The relationship between the radicalness score and these three features has been mapped using our hypothesis (as explained below). The overall proposed approach starts from cleaning and preprocessing the raw text followed by the extraction of these features and finally estimating the radicalness score for each of the online user using our hypothesis.

Before getting into the details of the methodology, we would like to present our hypothesis and the proposed text-related features being used in the calculation of this radicalness score. A detailed explanation of the methodology is also given in the following section.

The Hypothesis

After going through the literature, we have selected three features that have a direct proportionality relationship with the degree of radicalness in text. The three features are: (i) Similarity of text to domain, (ii) Classification of text as "Radical" or "Non-Radical" (or Radical Flag), and (iii) Sentiment of text. We hypothesize that radicalness or the degree of radicalization in an online user can be calculated using the similarity, radical flag and the sentiment present in the text posted by the social media user. Thus, the proposed hypothesis between radicalness score and these features can be stated as:

$$\text{Radicalness} \propto \text{Similarity to domain} \tag{1}$$

$$\text{Radicalness} \propto \text{Radical Flag} \tag{2}$$

$$\text{Radicalness} \propto \text{Sentiment} \tag{3}$$

Following section provides a detailed explanation of these features and the corresponding relationship with our proposed hypothesis.

1. **Similarity of text to a domain:**

Several existing research works (Kadriu, Abazi, and Abazi 2019; Text Similarities: Estimate the degree of similarity between two texts — by Adrien Sieg — Medium n.d.) suggest that similarity to a domain glossary can help in the classification of text into a particular category. The domain glossary comprises of frequently occurring words of the domain such as names of significant persons, places, dates, and events. In our approach, we propose to use similarity to the domain as a feature to estimate the radicalness score. The more similar the text is to the domain and its words, the more is the probability of text being related to the domain. This will help in eradicating any unrelated tweets or sentences not belonging to our domain of interest.

For the purpose of our research, we have considered a glossary of radical keywords (key names and locations) belonging to the conflict zones of Jammu and Kashmir. The tweets and online blogs have a plethora of radicalized words & phrases in this region as Jammu and Kashmir has been a hotspot of many terrorist events and attacks in the past few years. Thus, online radicalization is on a rise in this area. Further, we have classified the domain into three subdomains to measure the similarity at a more granular level. The subdomains are identified as Politics, Unrest, and crime. Separate glossaries of persons of interest and places of interest for Politics, Crime, and Unrest have been prepared in the proposed technique. This forms the first feature of our hypothesis.

Classification of text as "Radical" or "Non-Radical":

Next, we have used the" Radical Flag" or the" presence of radicalization" in given text as a feature in our hypothesis. The relationship between radicalness and radical flag is given in Eq. (2). According to the hypothesis, to determine the radicalness of a text we need to identify whether the text is radical or not. The radicalness will be less or zero for a non-radical text as compared to a highly radicalized text. Thus, it is important to find out the classification or the nature of the text. Several automated means of text classification can be found in the existing literature. In Kaur, Saini, and Bansal (2019) the authors have used LSTM as a technique to classify text into radical or non-radical. In the paper (Minaee et al. 2021), the authors have presented a comprehensive comparison of different text classification techniques using standard datasets such as Amazon reviews dataset and Yelp dataset. The model accuracy as high as 98% has been achieved by the deep learning models. Thus, suggesting that machine learning and deep learning-based models can prove to be effective text classification techniques. Taking motivation from the literature, we propose to find out the radical flag ("Radical" or "Non-Radical") of text using the CNN-LSTM classification algorithm. The performance and accuracy of our model is evaluated in section evaluation.

Sentiment of the text:

The last feature in our hypothesis is the sentiment of text. The relationship between the sentiment and the radicalness is stated in Eq. (3). As given in Eq. (3), we propose that there is a direct relationship between the sentiment of text and the radicalness. The more negative the sentiment in a sentence is, the more is the probability of the text being "Radical." A larger negative sentiment value suggests more radicalness in text. Similarly, a higher positive sentiment value suggests more non-radicalness in text. Similar observations have been given by authors in the existing literature. Several authors (Scrivens et al. 2015; Scrivens and Frank 2016) have suggested to use sentiment as a feature in determining the nature and polarity of text for detection of radical content. By taking motivation from literature, we have also utilized the sentiment information present in text as feature in our technique to measure the radicalness.

The Mathematical Representation

From the above discussion it can be concluded that radicalness score is a function of (i) Similarity to domain, (ii) Radical Flag, and (iii) Sentiment of

text. We can formulate a mathematical equation of these three features and our hypothesis given in section the hypothesis as follows:

Using constant of proportionality in Eq. (3),

$$\begin{aligned}\text{Radicalness Score} = k * \text{Similarity to domain} \\ * \text{Radical Flag} * \text{Sentiment}\end{aligned} \tag{4}$$

where k is the constant of proportionality.

For simplicity, we will take the "k" as 1

Further, we segregate the radicalness score into the radicalness scores of subdomains identified as Politics, Unrest, and Crime. For each of the subdomain,

$$\begin{aligned}\text{Political Radicalness Score} = \text{Political Similarity Score} \\ *\text{Political Radical Content Flag} \\ * \text{Political Radical Sentiment}\end{aligned} \tag{5}$$

$$\begin{aligned}\text{Unrest Radicalness Score} = \text{Unrest Similarity Score} \\ *\text{Unrest Radical Content Flag;} \\ * \text{Unrest Radical Sentiment}\end{aligned} \tag{6}$$

$$\begin{aligned}\text{Criminal Radicalness Score} = \text{Criminal Similarity Score} \\ *\text{Criminal Radical Content Flag} \\ * \text{Criminal Radical Sentiment}\end{aligned} \tag{7}$$

The Approach

This sections gives a detailed explanation of the approach used to calculate the radicalness score of user accounts. The proposed approach is depicted in Figure 1. The framework can be split into five steps:

1. Creation of glossary for domain
2. Calculation of similarity with glossary
3. Classification of text into radical or non-radical
4. Extraction of sentiment from text
5. Estimation of radicalness score

Each of these steps have been elaborated in the following sections.

Creation of Glossary for Domain

To estimate the first feature for our approach, i.e., Similarity to domain, we need to build a glossary of frequently occuring words (nouns) in our domain of interest. To create the glossary for the proposed approach, we use the content from online blogs and news articles from known radicalized websites. The dataset used in this article is taken from the dataset used

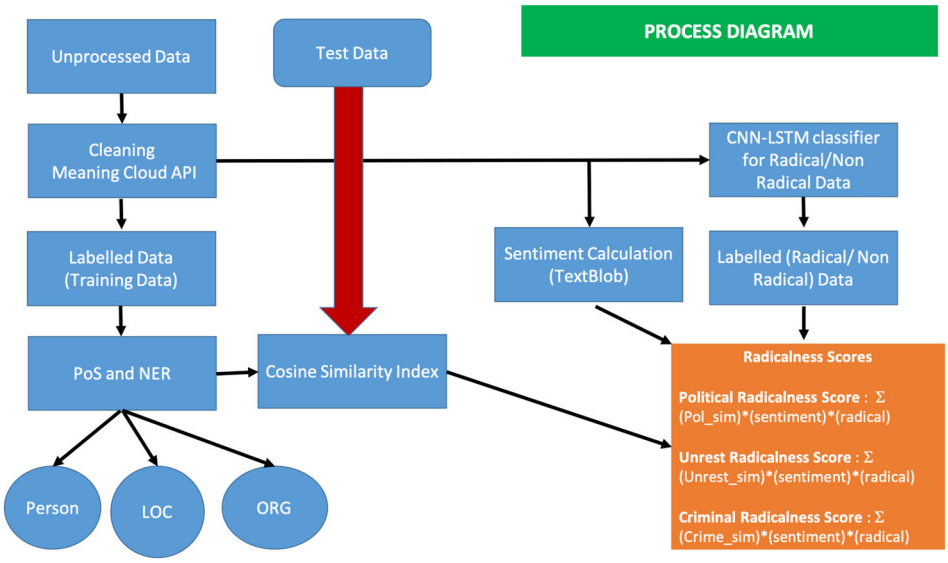


Figure 1. Framework for estimation of Radicalness Score.

Table 1. Dataset used in the proposed work.

Parameter	Value
Type of data	Online News, Articles, Blogs
Sources	Tribune, Greater Kashmir, Huriyat, JKLF World, Peoples League, Hizbul Muzahidin blogs
Time period	Several time windows between 2010 and 2018
Total size of training dataset	4.5 MB
Labels present in dataset	Radical, Non-Radical, Irrelevant

by the authors in paper (Kaur, Saini, and Bansal 2019). The description of the dataset used for creation of glossary are given in Table 1. The dataset has been labeled as "Radical," "Non-Radical," and "Irrelevant" by the annotators based on a predefined set of definitions for each of the labels. The process for data annotation has been explained in detail in section 3.1 of the paper (Kaur, Saini, and Bansal 2019). The dataset has been annotated by field experts for the presence or absence of radicalization in the text from Tweets, online blogs, and discussion forums using predefined definitions of radicalization. The authors have used the Kappa Coefficient to evaluate the agreement level of the annotators for the dataset used. These annotations are used for training as well as glossary creation. The distribution of labels is as follows:

- Radical Labels: 63% of total dataset
- Non-Radical Labels: 27% total dataset
- Irrelevant Labels: 10% of total dataset

(a) Glossary of Political Persons

(b) Glossary of Political Locations

(c) Glossary of Unrest Persons

(d) Glossary of Unrest Locations

(e) Glossary of Criminal Persons

(f) Glossary of Unrest Locations

Figure 2. Glossaries created for each subdomain.

The data obtained from these sources is then cleaned and preprocessed to remove all stop words, URLs, and alpha-numeric characters. The cleaned data is then sent to the MeaningCloud text classification API (Text classification and categorization API — MeaningCloud n.d.) for extracting the domain of the text. The MeaningCloud API uses the IPTC ontology to determine the category/domain of text. The domains to be considered in our proposed approach are: Politics, Unrest, and Crime. Once the domain or category of text is extracted, a Named Entity Recognition (NER) is performed over the text to extract the persons of interest and the locations of interest from the text. The person and place of interest are extracted for each of the subdomain (politics, unrest, and crime). Thus, we create a total

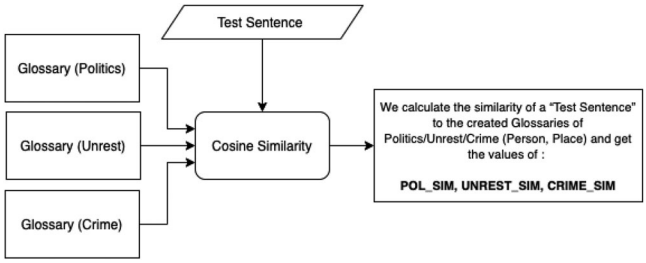


Figure 3. Calculation of similarity with domain words.

of six different glossaries with two subglossaries (person of interest and place of interest) each for the three subdomains (politics, unrest, crime). Rest of the content is categorized as miscellaneous. The glossaries created are represented in Figure 2 by means of a word cloud.

Calculation of Similarity with Domain Words

We now estimate the Similarity of text to domain words (glossary). We calculate the similarity of input text against the glossaries created in the previous step using cosine similarity. Cosine similarity is a popular NLP method for approximating how similar two word/sentence vectors are. The intuition behind cosine similarity is relatively straight forward. We simply use the cosine of the angle between the two vectors to quantify how similar two documents are. We use a python package (cosine_similarity) to calculate the similarity between text and the domain words of Politics, Unrest, and crime, respectively. Thus, we obtain three cosine similarity values: POL_SIM, UNREST_SIM, CRIME_SIM for each input text. The process is depicted in Figure 3. This forms the first feature for our approach.

Classification of Text

To find the next feature for our algorithm, we need to identify the radical flag of our text, i.e., classify the text as "Radical," "Non-Radical," or "Irrelevant." We have used only these categories for our approach as there is no standard nationally or internally known classification available w.r.t radicalization. This is an ongoing study and the categories will be refined in future research with the help of subject matter experts. For the purpose of classification, we propose to use the CNN-LSTM-based algorithm for determining the radicalization flag of the text. We have used the annotated data from the research work (Kaur, Saini, and Bansal 2019) to train and test our CNN-LSTM model. We have combined the advantages of two traditional neural network model, Long Short-Term Memory (LSTM) and Convolutional Neural Network (CNN) by effectively preserving the

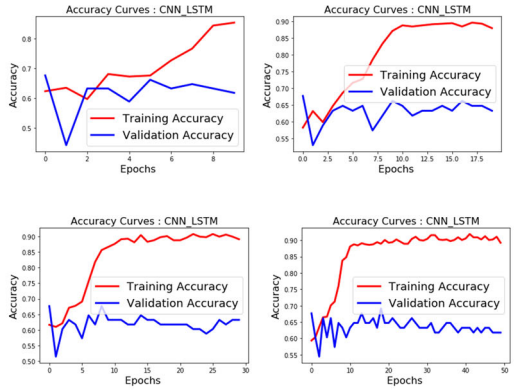
Table 2. Evaluation of CNN-LSTM classifier for radical text detection.

Validation split	No. of epochs	Training accuracy	Validation accuracy
0.1	10	86.42	61.76
	20	88.03	63.24
	30	89.65	63.24
	50	91.25	61.76
0.2	10	79.34	65.00
	20	91.31	66.67
	30	89.73	64.17
	50	92.21	64.17
0.3	10	72.23	67.52
	20	93.36	68.15
	30	91.63	66.24
	50	93.25	68.15

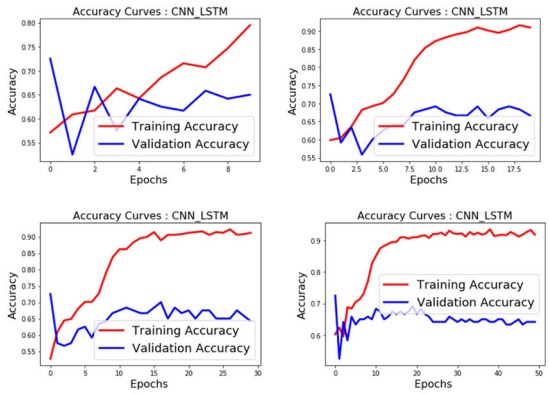
characteristics of historical information in long text sequences, and extracting local features of text by using the structure of CNN. Convolutional neural networks excel at learning the spatial structure in input data. This learned spatial features may then be learned as sequences by an LSTM layer. Thus, creating an effective sentence classification method for radicalization detection in text. The performance of the model used in our research work in terms of the training accuracy and validation accuracy is shown in Table 2.

It can be seen from Table 2 that the maximum training accuracy of the proposed CNN-LSTM classifier is around 93% at 20 epochs and a validation split of 0.3. As the number of epochs increase, there is no significant change in the training accuracy beyond 20 epochs whereas the validation accuracy seems to drop after the 20th epoch. Same can be observed in Figure 4.

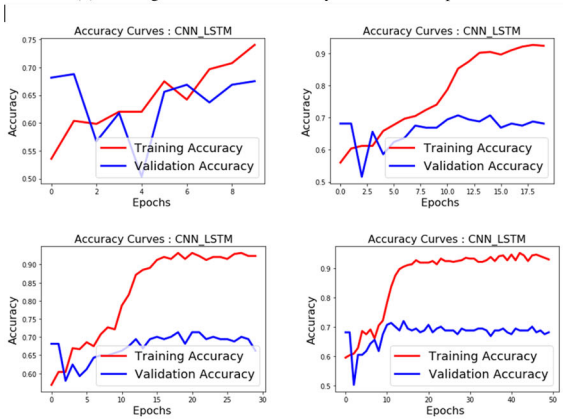
To compare the performance of the proposed CNN-LSTM classifier, we have applied several machine learning techniques to the same annotate dataset and the results have been presented in Table 3. It can be observed from our results that machine learning techniques like Logistic Regression, Support vector machine, Naïve Bayes, and Random forest are able to reach maximum accuracy of 60% whereas deep learning techniques like CNN, HAN, LSTM have been successful in achieving an accuracy of 63% to 91%. The Hierarchical Attention Network (HAN) is a deep learning model for document classification which is built from bidirectional RNNs composed of GRUs/LSTMs with attention mechanism. The HAN gives a classification accuracy of 91.78% at 20 epochs whereas CNN achieve 63.27% accuracy with 20 number of epochs. The CNN_LSTM classifier proposed in our current approach outperforms the other deep learning techniques and is able to achieve an accuracy of 93.36% in 20 epochs. Thus, suggesting that our CNN_LSTM-based classifier is a better radicalization detection technique for the social media text. The sample predicted labels after classification using our approach are shown in Table 4.



(a) Training and Validation accuracy for validation split of 0.1



(b) Training and Validation accuracy for validation split of 0.2



(c) Training and Validation accuracy for validation split of 0.3

Figure 4. Training and validation accuracy for CNN-LSTM classifier.

Extraction of Sentiment from Text

The next feature that we extract is the sentiment of the text, i.e., whether the sentence is positive, negative, or neutral. We have used the TextBlob python library (Tutorial: Quickstart – TextBlob 0.16.0 documentation n.d.)

Table 3. Evaluation of machine learning classifiers for radical text detection.

Technique	Accuracy (%)	Number of epocs
CNN	63.27	20
HAN	91.78	20
CNN_LSTM	93.36	20
LSTM	76.00	20
Linear SVC	60.29	
Logistic Regression	58.79	
Multinomial NB	55.68	
Random Forest Classifier	46.27	

Table 4. Predicted labels for dataset.

Predicted_Label	Text
Non-Radical	Cordial Relations the Veteran Kashmiri Leader, Syed Salahuddin, has Said That the Permanent Peace in South Asia
Radical	Blaming Coalition Regime for Kidnapping Hurriyat Leaders and Lodging Them at Unknown Destination in New Delhi, Syed Ali Geelani Called It Sheer Political Vendetta. Syed Ali Geelani Expressed His Deep Concern Over the Kidnapping of Ailing Hurriyat Leader, Peer Saif Ullah. Syed Ali Geelani While Lamenting At careless approach Of State Authorities NIA Said, they Stand Responsible for Their Negligence. Lashing at State Administration, Syed Ali Geelani Said That They Want to Gain Favors for These Arbitrary Actions from Their Masters in New Delhi &; Are Creating A Sense of Scare, However, We Will Fight Them Tooth and Nail, Added Geelani. They Won't Cow Down Our Passion and People in State Won't Deter from Their Mission and Will Pursue Their Fight For Freedom Right With Persistence

```
def analize_sentiment(tweet):
    '''
    Utility function to classify the polarity of a tweet
    using textblob.
    '''
    analysis = TextBlob(tweet)
    if analysis.sentiment.polarity > 0:
        return 1
    elif analysis.sentiment.polarity == 0:
        return 0
    else:
        return -1
```

Figure 5. Analysis of sentiment present in the text.

which is a Python (2 and 3) library for processing textual data. It provides a consistent API for diving into common natural language processing tasks such as part-of-speech tagging, noun phrase extraction, sentiment analysis, and more. The API returns a sentiment polarity value in between [−1.0,1.0] where a negative value represents a negative sentiment, and a positive value represents a positive sentiment. A zero value represents neutral sentiment. We have used algorithm given in Figure 5 to extract the sentiment of a sentence. The function analize_sentiment() will return "−1" or Negative as the aggregated sentiment value for all sentences for which sentiment polarity is less than zero. Similarly, it will return "1" or Positive as the aggregated sentiment value for all sentences for which sentiment polarity is greater than zero. A neutral sentiment polarity will be returned as "0" or

Table 5. Sample of sentences and corresponding aggregate sentiment.

Text	Predicted sentiment
Crazy Pakistan military wasted public money creating fake war	Negative
Great News India BJP government granted emergency powers to Indian Armed Forces to procure weapons	Positive
Pakistan turned to China	Neutral

Neutral. Example of aggregate sentence sentiments extracted using analize_sentiment() are shown in Table 5.

Estimation of Radicalness Score

The final step in our approach is to determine the radicalness score of each user. For this, we collect all the features together, i.e., Similarity of text to a domain, Classification of Text and Sentiment of Text and use the Eqs. (5)–(7) to calculate the radicalness score for each sentence in the radicalization dataset. To summarize, the final set of features that are used for estimating the radicalness score of online users are as follows:

- POL_SIM
- UNREST_SIM
- CRIME_SIM
- PREDICTED_FLAG (Radical, Non-Radical)
- SENTIMENT

This score is calculated for all sentences posted by the user on its account. Next, the individual sentence radicalness scores are summed up and collated using Eqs. (8)–(10) to estimate a final radicalness score for each user account. This is mathematically represented as follows:

$$\begin{aligned}\text{Political Radicalness Score} = \sum & \textit{P}\text{olical Similarity Score} \\ & * \text{Political Radical Content Flag} \\ & * \text{Political Radical Sentiment}\end{aligned} \tag{8}$$

$$\begin{aligned}\text{Unrest Radicalness Score} = \sum & \text{Unrest Similarity Score} \\ & * \text{Unrest Radical Content Flag} \\ & * \text{Unrest Radical Sentiment}\end{aligned} \tag{9}$$

$$\begin{aligned}\text{Criminal Radicalness Score} = \sum & \text{Criminal Similarity Score} \\ & * \text{Criminal Radical Content Flag} \\ & * \text{Criminal Radical Sentiment}\end{aligned} \tag{10}$$

Finally, we normalize the scores using mean and standard deviation and we get a Political, Unrest, and Criminal Radicalness score for each of the user accounts in the dataset used. These scores can be used to differentiate

between the originators and the propagators depending on the level of radicalness produced in the online content published by each user. Deciding the thresholds for the level of radicalness is beyond the scope of this article and can be decided by the security agencies based on field intelligence. The following section evaluates the effectiveness of the proposed technique using standard metrics of evaluation.

Evaluation

As described in the earlier sections, we have calculated the radicalness scores for the three categories: Politics, Crime, and Unrest using the proposed approach. This section discusses the results obtained from our analysis. The empirical results show that the observations are in accordance with the opinions of the national security forces and those found in literature.

Dataset for Evaluation

To carry out an evaluation of our proposed approach, user accounts were identified from the dataset that fulfilled the following criteria:

- Had produced more than 500 tweets in the given time window (2010 to 2018 and March–April 2019)
- Tweeted about topics related to Kashmir, Jihad, Surgical Strikes, anti-Indian content, content opposing the Indian Government, Tweets against the Indian Defence
- Or belonged to the Online blogs of known radicalized entities

Finally, a total of 20 users were identified which included 17 users from Twitter and three User blogs (treated as a single online entity). Some of the websites were blocked by the Indian government after the data had been extracted from these websites. The data from these websites, blogs, and twitter was multi-lingual with English, Urdu, and Arabic being the most common languages. The translation activity had to be undertaken to create sufficient data for the models. The account names have been masked due to confidentiality. Details of the user accounts are available in Table 6

Results and Discussion

The proposed algorithm was run on the online content of each of these users and corresponding Political, Unrest, and Criminal Radicalness scores were calculated. The estimated radicalness scores for each of these user

Table 6. Details of the blogs, user accounts used for evaluation.

User	User description
User_1	A known radicalized blog
User_2	News website of Pakistan
User_3	Local Kashmiri website with known radical content
User_4	A terrorist organization known to be working against the interests of India
User_5	Local Kashmiri website with known radical content
User_6	A terrorist organization known to be working against the interests of India
User_7 to User 20	Users from Twitter posting using identified keywords such as Kashmir.

Table 7. Political, unrest, and crime radicalness scores for each user.

Users	Political radicalness	Unrest radicalness	Crime radicalness	Total radicalness
User_1	−0.0001372	0.14703255	0.00186341	4.96 E − 02
User_2	0.00503007	0.0149899	0.00307816	7.70 E − 03
User_3	0.00194149	0.01957108	−0.0002679	7.08 E − 03
User_4	−0.0012388	0.02201459	−8.63 E − 06	6.92 E − 03
User_5	0.00455063	0.00031714	0.01268249	5.85 E − 03
User_6	0.0149142	0.00250107	−0.000672	5.58 E − 03
User_7	0.00203303	0.00121721	0.00392148	2.39 E − 03
User_8	0.00212718	−0.0001301	0.00274461	1.58 E − 03
User_9	0.00234574	−0.0001741	0.0022192	1.46 E − 03
User_10	−0.0007283	0.00029513	0.00259066	7.19 E − 04
User_11	0.00089337	−0.0020052	0.00314615	6.78 E − 04
User_12	0.00109638	−0.0041436	0.00373683	2.30 E − 04
User_13	0.00026785	−0.0008743	0.00074481	4.61 E − 05
User_14	−9.80 E − 05	−0.0004362	0.00020007	−1.11 E − 04
User_15	−9.82 E − 05	−0.0007594	0.00020032	−2.19 E − 04
User_16	−0.0004781	−0.0018072	0.00103197	−4.18 E − 04
User_17	−0.0004348	−0.0015399	0.00020704	−5.89 E − 04
User_18	−0.0004342	−0.0015413	0.00020675	−5.90 E − 04
User_19	0.00012938	−0.0024386	5.22 E − 05	−7.52 E − 04
User_20	0.00117648	−0.0060317	0.00039166	−1.49 E − 03

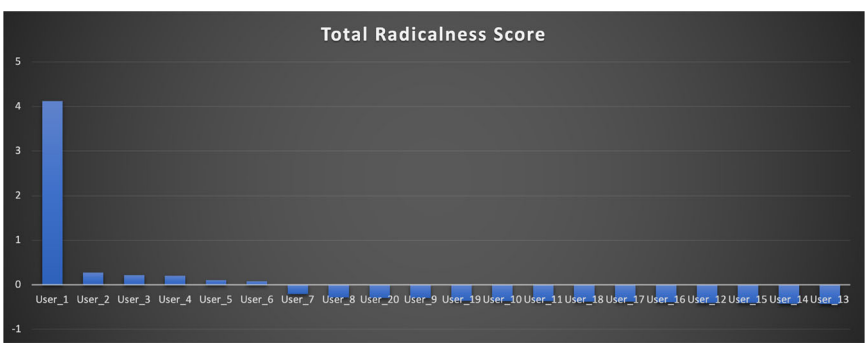


Figure 6. Radicalness scores of all users.

accounts is presented in Table 7. A graphical representation of the radicalness scores for these user accounts is shown in Figure 6.

It can be observed in Figures 6 and 7 that six users User 1 till User 6 depict higher values of total radicalness score than other accounts where the radicalness scores are much less. This shows that these six accounts produce content that is much more extreme in nature than the other

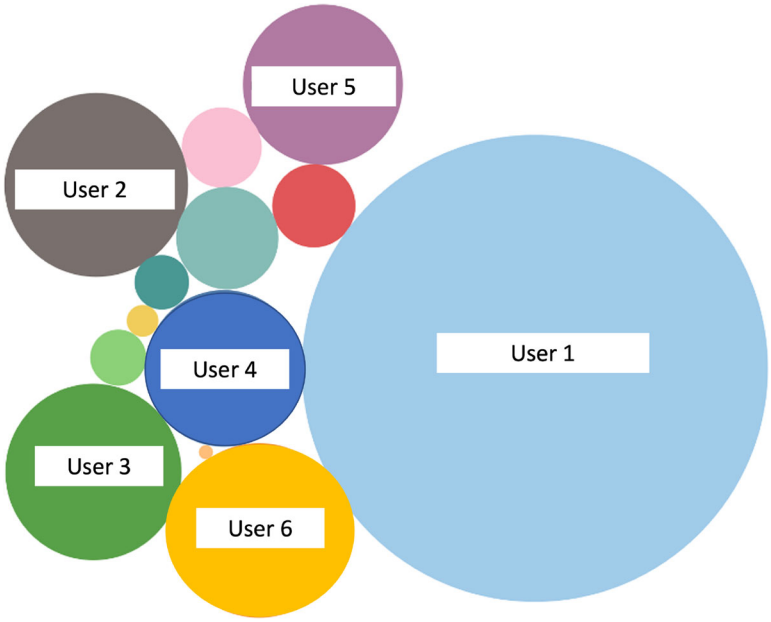


Figure 7. Total Radicalness scores of all users.

accounts. It can be noted here that the total radicalness score of User 1 blog is almost 300% deviated from the mean of total radicalness of all user accounts put together. Thus, empirically proving a visible radical activity in the content belonging to these blogs. A notable point here is that, out of these six accounts three accounts have been crawled from two restricted websites of a terrorist organization known to be working against the interests of India and one radicalized blogging URL of an organization that has been posting a lot of content against the Indian armed forces and government in the name of supporting a Free Kashmir. The other three user accounts belong to a news website of Pakistan and two local Kashmiri online news accounts. It is a known fact that some local newspapers and some radicalized news from Pakistan propagate anti-Indian content over social media.

To further verify our claim, we have compared the individual normalized political radicalness scores of the user accounts. Figure 8 shows the observations of this analysis. It is observed that User 6 has the highest political radicalness score compared to others. Political Radicalness score of a User 6 Blog is 88.90% deviated from the mean of all other user accounts out together. This implies that User 6 is a blog that uses a lot of political words such as names and locations of the Indian Government with a very high negative sentiment and the nature of content being very radical. This has been observed in a manual inspection of all blogs of User 6. We can safely assume our analysis to be accurate.

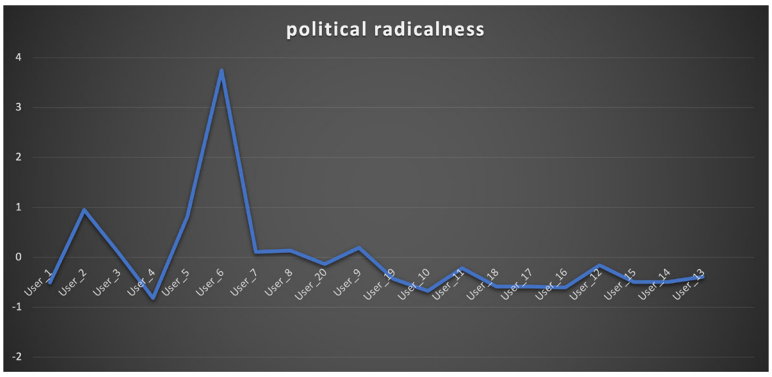


Figure 8. Comparison of Political Radicalness scores of all users.

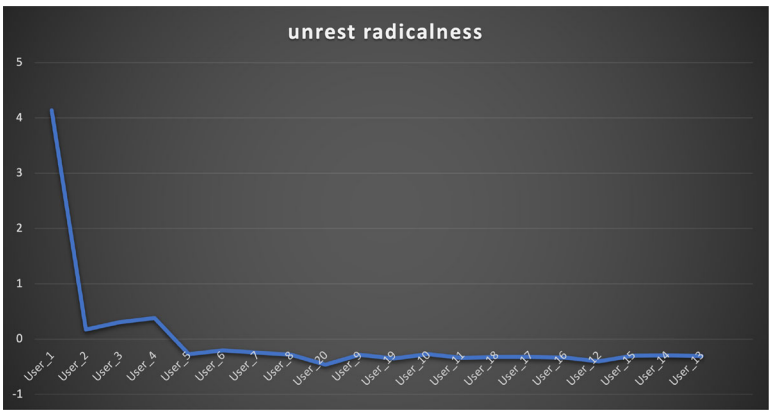


Figure 9. Comparison of Unrest Radicalness scores of all users.

Next, we have analyzed the unrest radicalness normalized score of all the user accounts. Figure 9 shows the observations of this analysis. It is observed that User 1 has the highest unrest radicalness score compared to others. Unrest Radicalness of a User 1 Blog is 14.80 times deviated from the mean of all other user accounts out together. This implies that User 1 is a blog that uses a lot of unrest/war related words such as names and locations of the Terrorists, war terminology with a very high negative sentiment and the nature of content being very radical. This has been observed in a manual inspection of all blogs of User 1. We can safely assume our analysis to be accurate.

Next, we have analyzed the criminal radicalness normalized score of all the user accounts. Figure 10 shows the observations of this analysis. It is observed that User 5 (a local Kashmiri website) has the highest crime radicalness score compared to others. Crime Radicalness of a User 5 is highly deviated from the mean of all other user accounts out together. This implies that User 5 is a blog that uses a lot of crime related words such as murder, homicide etc. with a very high negative sentiment and the nature

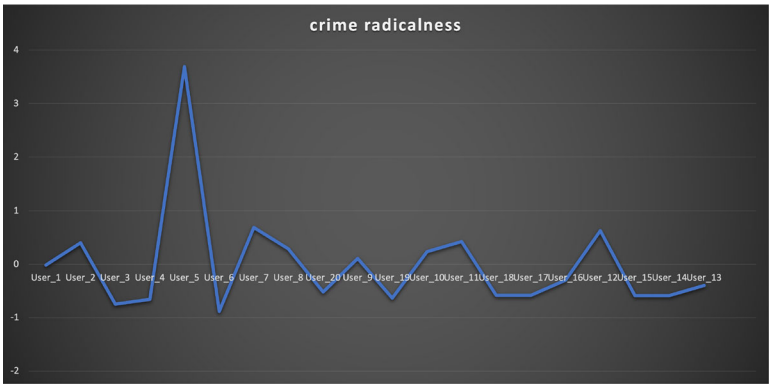


Figure 10. Comparison of Crime Radicalness scores of all users.

of content being very radical. This has been observed in a manual inspection of all blogs of User 5. We can safely assume our analysis to be accurate.

These scores can be used to identify the propagators as well as the originators of radical content. The thresholds can be decided as per the requirements of the authorities since each authority may be interested in finding out a particular set of radicalized people. Deciding the threshold is beyond the scope of this article.

Final Discussion and Conclusion

In this study, we have presented an approach to calculate the radicalness score for the online user accounts that will help to measure the amount of radicalization being propagated by them. The estimation of the degree of radicalization can help LEA to track down the originators as well as the propagators of radicalization. The accounts with very high radical score can be suspended or put to tracking for suspicious activities. Thus, reducing the manual effort of scrapping through all user accounts. Also, a major contribution of our approach is the segregation of radicalization into categories like politics, unrest, and crime. Each type of radicalization can attract different types of penalties and treatment. Thus, enforcing law and order in a more systematic way. De-radicalization efforts will also be much strengthened by introducing a data driven radicalness scoring approach.

This scoring approach has been designed using three important text-based features which are identified as Similarity to domain, the sentiment present in the text and whether the text is classified as radical or not. These are key features used to understand the intent conveyed in any text. Each of the features is extracted carefully using State of the art techniques such as cosine similarity, Text blob sentiment analysis and CNN-LSTM classifier respectively. We have used Meaning cloud API and NER for

creating the domain specific glossaries for Politics, Crime, and Unrest subdomains. Both the coding libraries are widely used and have been developed using Neural Network-based models. We have proposed a CNN-LSTM-based text classifier for classification of text into "Radical" or "Non-Radical." The accuracy achieved on our dataset is around 93%. This is a significant enhancement in results in comparison to the existing ML and AI techniques. The empirical results of our proposed approach resonate with the observations of the existing literature. For instance, our results show that the contents from a terrorist website has a larger political radicalness score compared to other accounts. This can also be verified from the text of the same website where we can find the mention of several Indian political people and places. Similar observations can be made using the other empirical results of our approach. To conclude, we can summarize the major contributions of our paper as below:

- Creation of an approach for estimating the radicalness score of Online Social media accounts
- Mathematical and scientific definition of radicalness score based on measured features present in online social media text. The features have been verified and supported by the literature studied for this research work.
- CNN-LSTM-based technique for detection of radicalization in text that has achieved an accuracy of around 93% and is much computationally effective than other deep learning models.

To the best of our knowledge, our research is the first of its kind attempt to identify and quantify radicalization of online users. We believe this is a novel contribution of our work and we have not come across such research in our survey of the exiting literature. A valuable outcome of our approach is an automated mechanism for radicalization detection and quantification. This can certainly help the security agencies to make the process of identification of radical agents more robust and effective. This will reduce the effort that goes in manually identifying radicalized online accounts and the human error that may seep in. Thus, an automatic radicalization detection and quantification mechanism using AI and ML will strengthen the security forces to a large extent and create more opportunities for research and development in this area.

Future Directions

The radicalness score calculated in this article is a significant step toward identifying the radical agents that disrupt the functioning of the society.

The quantification of the radicalness in a user account can help identify the influence and potential of that user to radicalize its followers. Security agencies can use this score to take legal action or setup a mitigation plan to bring back the disillusioned youngsters. Radicalization of the youth is not only be a big concern for the security agencies but also impacts the lives of citizens. Frequent gun shots in the United States by young men/women is a huge security breach and in most cases results from the extreme radicalization and discontent. According to Wikipedia, in the year 2021, a total of 693 mass shootings occurred in United States killing 703 people and injuring 2842. A detailed analysis into the Social Media accounts of the shooters can indicate the radical behavior of these users. Similarly, misinterpreted religious teachings propagated by religious gurus can lead to disharmony and mobilization of people. If such a content floats online without regulation, the results can be disastrous. Hence, our scoring mechanism can serve a powerful tool to keep a check on such text and prevent radicalization.

There are several other research directions that can be chalked out of the current study. For instance, the authors in the paper (Uba and Bosi 2022) talk about the socio-economic factors that lead to radicalization. These features can be considered to improve the radicalness score further. Demographic factors like user location, gender, age, likes, dislikes, and browser history can be incorporated in the calculation of radicalness score. Machine Learning and Deep Learning techniques can be further strengthened using concrete dictionaries and domain glossaries. To conclude, it can be stated that the proposed radicalness score can help in effectively detecting and quantifying elements of radicalization on social media.

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